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Light verbs, full verbs:

The same verb can occur in a **light verb construction LVC** (*make a decision*) or as a **full lexical verb** (*make a cake*). In these constructions the verb looks alike, while its lexical and semantic properties differ.

Light verb: The verb contributes little lexical meaning; the noun carries the main semantic content (*decision* denotes the act of deciding).
Full verb: The verb retains its lexical meaning and contributes centrally to the event described (*make* denotes creating something).

Can language models capture this distinction under controlled syntactic and contextual conditions?

- RQ1:** Do LMs assign different probabilities to light- and full-verb constructions?
RQ2: Is this distinction reflected in contextual embeddings?
RQ3: How consistent are these patterns across languages and models?

Dataset

Contrast minimal series of LVC and full verbs

Verbs are paired with different objects, resulting in either light-verb or full-verb uses.

Languages: English · French · Spanish
5 high frequency verbs per language:
make, take, give, have, receive
faire, prendre, donner, recevoir
hacer, tomar, dar, tener, recibir

Verb	Light-verb use	Full-verb use
make	make a decision	make a cake
take	take a walk	take a mug
have	have a celebration	have a car
give	give a look	give a coin
receive	receive attention	receive a package

Expressions are inserted in matched sentence contexts with
Variations in temporal and spatial specifications,
Variations in tense, subjects, and other grammatical properties
including **active and passive voice**

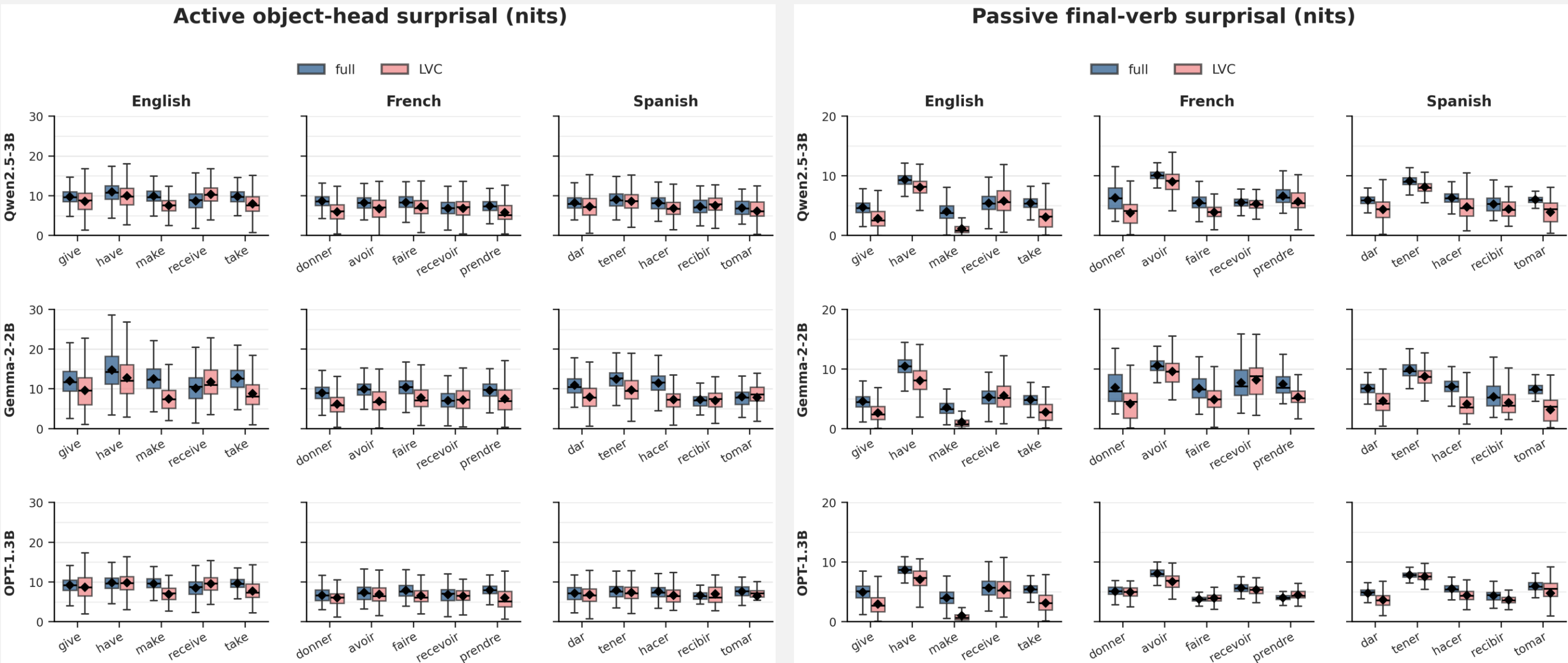
Condition	Example
LVC active	During the day, they made a decision.
Full-verb active	During the day, they made a cake.
LVC passive	During the day, a decision was made.
Full verb passive	During the day, a cake was made.

Surprisal-based probing

Does light-verb status affect word predictability? Surprisal measures how unexpected a word is in a context $S(w_t) = -\log P(w_t \mid w_{<t})$

Active: Measure surprisal of the object noun following the verb (47,600 sentences from the dataset)
Passive: Measure surprisal of the verb following the nominal object (47,600 sentences from the dataset)

LVCs generally have lower surprisal than matched full-verb constructions across six tested language models and three languages.

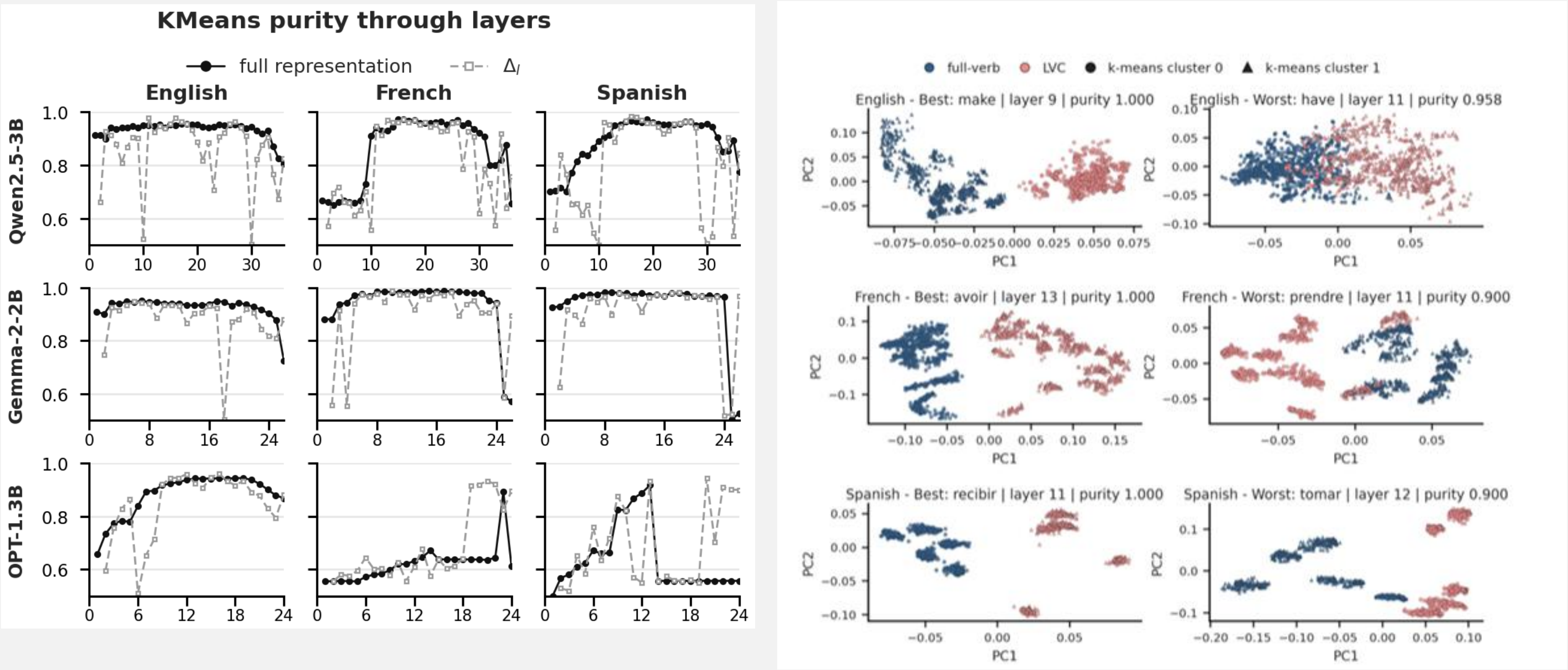


Contextual Embedding Analysis

Do LVCs and full verbs occupy different regions in the representation space?
We extract contextual representations of object noun phrases and compare them using dimensionality reduction and clustering.

LVC and full-verb objects show clear separation across models and layers.

Separation is more stable in accumulated representations than in layer-to-layer changes. The strength of separation varies across models and languages, and semantic properties of nouns can contribute to it.



Limitations/further research

The database is being expanded to more languages ad more naturalistic contexts. Semantic properties like noun concreteness may contribute to embedding separation, independently of light-verb status.

Conclusions

The proposed multilingual dataset provides a controlled framework for investigating phraseological competence, revealing that language models are sensitive to the light-verb/full-verb distinction in both their predictions and contextual representations.



Full dataset and sentence generation code
Available at:
github.com/XplainLing/LVC_sentences_database



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